PECH GROUP HOLDINGS LTD

Technology & Infrastructure Enablers for Africa

CRM & ERP Integration Research for PECH Group Holdings Ltd

Table of Contents

Table of Contents

1. What is CRM
2. CRM vs ERP — How They Differ and Complement Each Other
3. The PECH Business Architecture
4. CRM Market Size — Global, Africa, Nigeria
5. Open-Source CRM Options (License-Safe)
6. PECH Recommended CRM Stack
7. CRM Modules & Features for African Businesses
8. CRM-ERP Integration Architecture
9. Mobile-First & USSD CRM for Africa
10. CRM + AI Integration
11. Integration with PECH Platform Stack
12. CRM Data Model
13. Implementation Roadmap
14. Key CRM Metrics & KPIs
15. NDPA 2023 Compliance
16. Cost Analysis — Self-Hosted CRM in Lagos
17. Sources

1. What is CRM

Customer Relationship Management (CRM) is a strategy, process, and technology for managing all of a company's interactions with current and potential customers. A CRM system centralizes customer data, tracks interactions across touchpoints (sales, marketing, support), and automates workflows to improve customer retention, drive sales, and increase profitability.

Core CRM Functions

Function	Description
Contact Management	Centralized database of all customers, leads, and prospects with full interaction history
Lead Management	Capture, qualify, score, assign, and nurture leads through the sales funnel
Opportunity/Deal Tracking	Track deals through pipeline stages from prospect to close

Sales Automation	Automate repetitive sales tasks — follow-ups, reminders, quote generation
Marketing Automation	Email campaigns, customer segmentation, campaign ROI tracking
Customer Service	Ticketing, case management, SLA tracking, knowledge base
Analytics & Reporting	Dashboards, forecasts, pipeline reports, conversion metrics
Communication Tracking	Email sync, call logs, meeting notes, WhatsApp messages

Why CRM Matters for PECH

PECH operates across **12 business domains** (IoT, solar, commerce, fintech, etc.). Without CRM: - Customer data is siloed across departments - Sales teams can't track which IoT prospects also need solar solutions - Cross-sell and upsell opportunities are lost - Customer support lacks context from sales conversations - No unified view of customer lifetime value across product lines

---

2. CRM vs ERP

Key Differences

Dimension	CRM	ERP
Focus	External — customers, prospects, relationships	Internal — operations, resources, processes
Primary Users	Sales, marketing, customer service	Finance, operations, HR, procurement
Core Data	Contacts, leads, deals, interactions	Invoices, inventory, payroll, manufacturing
Goal	Increase revenue through better customer relationships	Reduce costs through operational efficiency
Scope	Front-office	Back-office
Revenue Impact	Drives top-line growth	Optimizes bottom-line costs

How They Complement Each Other

```
Customer Journey:

  [Marketing]  →  [Sales]  →  [Order]  →  [Fulfillment]  →  [Support]
      ↑              ↑           ↑              ↑               ↑
    CRM            CRM        CRM+ERP          ERP             CRM
                              (handoff)
```

The CRM-ERP handoff is critical. When a deal closes in CRM: 1. CRM creates the customer record 2. ERP creates the sales order, invoice, and delivery 3. CRM tracks post-sale satisfaction and support 4. ERP tracks payment, inventory depletion, and cost

Without integration: Sales teams don't know if inventory is available. Finance doesn't know what deals are closing. Support doesn't know the customer's purchase history.

ERPNext's Built-In CRM Module

ERPNext (MIT license, already in PECH stack) includes a CRM module with: - Lead management with auto-assignment rules - Opportunity tracking with pipeline visualization - Lead scoring based on engagement - Quotation generation linked to inventory - Campaign tracking - Integration with Accounting, Sales, and Inventory modules

This is significant for PECH — ERPNext's CRM module may cover 60-80% of needs without adding another system.

3. The PECH Business Architecture

The full PECH business stack follows a layered architecture where CRM sits at the customer-facing layer:

```
Mobile Apps / Websites / USSD
        ↓
Customer Layer (CRM)                    ← Customer-facing interactions
        ↓
Marketplace Layer (OMS / Medusa)        ← Commerce & orders
        ↓
Operations Layer (ERP / ERPNext)        ← Internal operations
        ↓
Supply Chain Layer (SCM + WMS)          ← Logistics & inventory
        ↓
Financial Infrastructure (Fineract)     ← Payments, lending, banking
```

How CRM Fits Each Layer

Layer	CRM Role	Integration Point
Customer Layer	Primary CRM — lead capture, deal tracking, customer profiles	Keycloak (SSO), Chatwoot (support)
Marketplace Layer	Customer purchase history, order status visibility	Medusa (commerce), order management
Operations Layer	Quote-to-cash flow, customer-linked invoices	ERPNext (accounting, inventory)
Supply Chain Layer	Delivery status for customers, SLA tracking	ERPNext (warehouse, delivery)
Financial Layer	Customer credit scoring, payment history, loan status	Apache Fineract (lending, savings)

Cross-Domain CRM Value for PECH

A customer who buys a **solar panel** (Energy domain) may also need: - **IoT monitoring** (ThingsBoard) for their solar system - **Financing** (Fineract) for the purchase - **Smart home devices** (Commerce/Medusa) for their property - **Technical support** (Chatwoot) for installation - **Maintenance contracts** (ERPNext) for ongoing service

CRM provides the **360-degree view** that connects all these touchpoints into one customer record.

4. CRM Market Size

Global CRM Market

Metric	Value
Market Size (2025)	$112.91 billion
Market Size (2026)	$126.17 billion
Projected (2034)	$320.99 billion
CAGR	12.40%
Cloud-based deployments	87%

Adoption (10+ employees)	91%

CRM is the **largest and fastest-growing enterprise software category** globally.

Africa CRM Market

Metric	Value
Market Size (2025)	~$3.8 billion
Projected (2029)	$1.94 billion (Statista software-only)
Projected (2033)	$10.05 billion (broader market)
CAGR	12.9%
Global Share	3.8%
Growth Rate	Highest globally (with MEA)

Key drivers: digital transformation, expanding retail/banking sectors, growing middle class, increased foreign tech investment.

Nigeria CRM Market

Metric	Value
Market Size (2025)	~$163 million
Global CRM Share	1.69%
Rank in Africa	#2 (after South Africa)
Key Sectors	Fintech, telecoms, retail
Growth Driver	Mobile-first consumers, SME expansion

Zoho Nigeria achieved 70% YoY customer growth by pricing below global rates and offering locally relevant features. This validates the opportunity for affordable, self-hosted CRM solutions in Nigeria.

---

5. Open-Source CRM Options (License-Safe)

License Compatibility Matrix

PECH requires **Apache-2.0 or MIT** licensed software. Here's how the major open-source CRMs stack up:

CRM	License	Compatible?	Database	Language	Notes
ERPNext CRM Module	GPL-3.0	**Already in stack**	PostgreSQL/MariaDB	Python (Frappe)	Built-in, no separate install
Frappe CRM	AGPL-3.0	**NO**	PostgreSQL/MariaDB	Python (Frappe)	Excellent but AGPL
NocoBase	**Apache-2.0**	**YES**	PostgreSQL, MySQL, SQLite	TypeScript/Node.js	No-code CRM builder
BottleCRM (Django-CRM)	**MIT**	**YES**	PostgreSQL	Python/Django + SvelteKit	Lightweight, startup-focused
Krayin CRM	**MIT**	**YES**	MySQL/PostgreSQL	PHP/Laravel + Vue.js	Full-featured, SME-focused
Twenty CRM	GPL-3.0	**NO**	PostgreSQL	TypeScript	Modern UI but GPL
SuiteCRM	AGPL-3.0	**NO**	MySQL/MariaDB	PHP	Enterprise-grade but

					AGPL
EspoCRM	AGPL-3.0	**NO**	MySQL	PHP	Lightweight but AGPL
Odoo CRM	LGPL-3.0	**NO**	PostgreSQL	Python	Huge ecosystem but LGPL

License-Safe Recommended Options

Tier 1: ERPNext CRM Module (Already Available)

- **License:** GPL-3.0 (already accepted into PECH stack)
- **Pros:** Zero additional deployment, deeply integrated with accounting/inventory/HR, Python/Frappe ecosystem
- **Cons:** CRM UX not as polished as dedicated CRM tools, limited marketing automation
- **Verdict:** Start here. Cover 60-80% of CRM needs with zero new infrastructure.

Tier 2: NocoBase (Apache-2.0) — Build Custom CRM

- **License:** Apache-2.0 (fully compatible)
- **Pros:** No-code CRM builder, PostgreSQL native, AI agent integration, plugin architecture, highly customizable
- **Cons:** Requires CRM configuration (not pre-built), smaller community than established CRMs
- **Best for:** Building a tailored PECH CRM that exactly fits the multi-domain business model
- **GitHub:** 15,000+ stars, active development

Tier 3: Krayin CRM (MIT) — Pre-Built Full CRM

- **License:** MIT (fully compatible)
- **Pros:** Full CRM out of the box, pipeline management, lead scoring, WhatsApp extension, multi-tenant SaaS mode, VoIP integration
- **Cons:** PHP/Laravel stack (PECH stack is mainly Python), MySQL-preferred
- **Best for:** If you need a complete CRM immediately without building from scratch

Tier 4: BottleCRM / Django-CRM (MIT) — Lightweight Python CRM

- **License:** MIT (fully compatible)
- **Pros:** Python/Django (aligns with PECH Python ecosystem), PostgreSQL with Row-Level Security, Flutter mobile app, SvelteKit frontend
- **Cons:** Smaller feature set, smaller community, startup-focused
- **Best for:** If you want a Python-native CRM that's easy to extend

---

6. PECH Recommended CRM Stack

Recommended Strategy: ERPNext CRM + NocoBase Custom Extensions

```
Phase 1 (Crawl): ERPNext CRM Module
   └── Leads, Opportunities, Quotations, basic pipeline
   └── Already deployed, zero additional cost
   └── Covers: Sales team of 5-10 reps

Phase 2 (Walk): Add NocoBase for Custom CRM Views
   └── Custom dashboards for multi-domain customer view
   └── AI-powered lead scoring (connect to Qwen2.5)
   └── WhatsApp/USSD integration layer
   └── Mobile-optimized CRM interface
   └── Covers: 20+ users, cross-domain selling
```

```
Phase 3 (Run): Full CRM Platform
   └── NocoBase as primary CRM UI
   └── ERPNext as back-office ERP
   └── Event-driven sync via NATS/Kafka
   └── AI agents for automated follow-ups
   └── Predictive analytics with XGBoost/LightGBM
   └── Covers: Full team, all 12 domains
```

Why This Combination Works

Factor	ERPNext CRM	NocoBase CRM
License	GPL-3.0 (accepted)	Apache-2.0
Database	PostgreSQL	PostgreSQL
Deployment	Docker	Docker
API	REST	REST + GraphQL
Customization	Frappe framework	No-code + plugins
AI Integration	Limited	Built-in AI agents
Mobile	Responsive web	Responsive web

---

7. CRM Modules and Features for African Businesses

Essential CRM Modules

7.1 Contact & Account Management

- Unified customer database across all PECH domains
- Company + individual contact hierarchy
- Custom fields for Nigeria-specific data (BVN reference, NIN, business RC number)
- Multi-currency support (NGN, USD, GBP, EUR)
- Multi-language profiles (English, Yoruba, Hausa, Igbo)

7.2 Lead Management

- Multi-channel lead capture: website forms, WhatsApp, USSD, phone calls, trade shows, referrals
- Lead source tracking (critical for ROI analysis)
- Auto-assignment rules: by territory (Lagos, Abuja, Port Harcourt), by product domain, by sales rep capacity
- Lead scoring based on: engagement, budget indicators, company size, urgency signals
- Lead nurturing workflows with automated follow-ups

7.3 Sales Pipeline

- Visual Kanban board (drag-and-drop deal stages)
- Customizable pipeline stages per business domain:
- **Solar:** Inquiry → Site Survey → Proposal → Negotiation → Installation → Commissioning
- **IoT:** Inquiry → Assessment → PoC → Proposal → Deployment → Go-Live
- **Commerce:** Browse → Cart → Checkout → Delivery → Review
- Win/loss analysis with reason tracking
- Revenue forecasting by pipeline stage

7.4 Quotation & Proposal Management

- Template-based quotations linked to ERPNext inventory
- Multi-currency pricing (auto-convert NGN ↔ USD)
- Digital signature / approval workflows
- PDF generation with PECH branding
- Version tracking (quote v1, v2, v3)

7.5 Marketing Automation

- Email campaigns with segmentation
- WhatsApp broadcast campaigns (critical for Nigeria)
- SMS/USSD campaign integration
- Campaign ROI tracking
- Customer segmentation: by domain, geography, spend level, engagement

7.6 Customer Service Integration

- Linked to Chatwoot for support ticketing
- SLA tracking per customer tier
- Escalation workflows
- Customer satisfaction (CSAT) surveys
- Knowledge base for self-service

7.7 Analytics & Dashboards

- Sales performance dashboards (Apache Superset integration)
- Pipeline health metrics
- Lead conversion funnels
- Revenue attribution by channel/campaign
- Customer lifetime value (CLV) analysis
- Geographic heat maps (Lagos, Abuja, PH, Ibadan)

Nigeria-Specific Features

Feature	Why It Matters
WhatsApp integration	90M+ Nigerian WhatsApp users; primary business communication channel
USSD fallback	40%+ of Nigerians use feature phones; USSD reaches everyone
Multi-currency	NGN volatility requires USD benchmarking for B2B deals
Offline mode	Intermittent internet; mobile CRM must work offline and sync
Agent network support	Field agents in secondary cities need mobile-first CRM
PAYG/installment tracking	Pay-as-you-go models are essential for solar and IoT adoption
BVN/NIN integration	KYC requirements for fintech-linked customers

---

8. CRM-ERP Integration Architecture

Integration Patterns

Pattern 1: Shared Database (Phase 1 — Simplest)

```
ERPNext CRM ←→ ERPNext ERP
    (same PostgreSQL database)
```

- Zero integration needed — they're the same system
- Best for: Phase 1 (Crawl) with small team

Pattern 2: Event-Driven Integration (Phase 2 — Recommended)

```
NocoBase CRM  →  NATS  →  ERPNext ERP
     ↓               ↓            ↓
 Lead Created   Event Bus   Create Customer
 Deal Won       Event Bus   Create Sales Order
 Quote Sent     Event Bus   Reserve Inventory
```

- Decoupled, scalable, fault-tolerant
- Uses NATS (Phase 1) → Kafka (Phase 3) for event streaming
- Best for: Phase 2+ with multiple systems

Pattern 3: API Gateway Integration (Phase 3 — Enterprise)

```
                 Apache APISIX
                      ↓
   ┌──────────────────────────────────┐
   │          API Gateway Layer       │
   │  Rate limiting, auth, routing    │
   └──────────────────────────────────┘

       ↓          ↓          ↓          ↓
    NocoBase   ERPNext    Medusa    Fineract
     (CRM)      (ERP)   (Commerce) (Fintech)
```

- All systems communicate through Apache APISIX
- Keycloak provides unified authentication
- Best for: Phase 3 with full microservices

Key Data Flows

CRM Event	ERP Action	Data Exchanged
Lead converted to customer	Create Customer record	Name, contact, address, tax ID
Deal won	Create Sales Order	Products, qty, price, terms
Quote approved	Create Quotation	Line items, discounts, taxes
Customer complaint	Create Issue	Ticket details, priority, SLA
Payment received	Update Invoice	Amount, date, payment method
Product delivered	Update Delivery Note	Delivery date, serial numbers

Data Synchronization Best Practices

1. **Single source of truth:** CRM owns customer data, ERP owns financial data
2. **Bi-directional sync:** Changes in either system propagate via events
3. **Conflict resolution:** Last-write-wins with audit trail
4. **Data validation:** Standardize formats at the API gateway level
5. **Idempotent operations:** Ensure duplicate events don't create duplicate records

---

9. Mobile-First and USSD CRM for Africa

Mobile CRM Architecture

```
┌──────────────────────────────────────┐
│          Customer Touchpoints        │
├──────────────┬──────────────┬────────┤
│ WhatsApp     │ USSD         │ Mobile App │
│ (90M         │ (Feature     │ (Smart-    │
│  users)      │  phones)     │  phones)   │
└──────────────┴──────────────┴────────┘
      │            │            │
      ▼            ▼            ▼
┌──────────────────────────────────────┐
│      Communication Gateway           │
│   (Chatwoot + Africa's Talking)      │
└──────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────┐
│          CRM Core (NocoBase)         │
│   Contacts, Leads, Deals, Reports    │
└──────────────────────────────────────┘
                │
                ▼
┌──────────────────────────────────────┐
│        ERP Backend (ERPNext)         │
│   Orders, Inventory, Accounting      │
└──────────────────────────────────────┘
```

WhatsApp CRM Integration

WhatsApp is the **#1 business communication channel** in Nigeria. CRM-WhatsApp integration enables:

- **Inbound lead capture:** Customer messages create CRM leads automatically
- **Automated responses:** Greeting, FAQ, business hours, product catalogs
- **Deal updates:** Sales reps send quotes and proposals via WhatsApp
- **Payment reminders:** Automated PAYG payment reminders
- **Customer surveys:** Post-service CSAT surveys via WhatsApp
- **Broadcast campaigns:** Product launches, promotions, event invites

Implementation: Chatwoot (already in PECH stack) supports WhatsApp Business API integration. Connect Chatwoot → CRM via API/webhooks.

USSD CRM Fallback

For the 40%+ of Nigerians on feature phones:

```
USSD Menu Structure:
*123#
├── 1. Check Order Status
│   └── Enter Order ID → Shows status
├── 2. Request Callback
│   └── Creates CRM lead with phone number
├── 3. Make Payment
│   └── USSD payment → Updates CRM + Fineract
├── 4. Report Issue
│   └── Creates support ticket in CRM
├── 5. Product Catalog
│   └── Browse products via text menus
└── 0. Speak to Agent
    └── Connects to Chatwoot support
```

Implementation Options: - **Africa's Talking USSD API** — programmatic USSD for Nigeria, Kenya, Uganda, Rwanda - **Celcom Africa** — USSD services across 10+ African countries - **HelloDuty** — CRM-

integrated USSD platform from Kenya

Offline-First Mobile CRM

For field agents in areas with poor connectivity:

1. **Local SQLite database** on mobile device
2. **Queue-based sync** — actions queued offline, synced when online
3. **Conflict resolution** — server timestamp wins for concurrent edits
4. **Progressive Web App (PWA)** — works offline via service workers
5. **Low-bandwidth mode** — text-only, compressed images, minimal data transfer

---

10. CRM AI Integration

AI-Powered CRM Features (Self-Hosted with PECH Models)

AI Feature	Model	Use Case
Lead Scoring	XGBoost / LightGBM	Predict lead-to-customer conversion probability
Sales Forecasting	Prophet / LightGBM	Predict monthly/quarterly revenue by pipeline
Email Generation	Qwen2.5-7B	Draft personalized follow-up emails
Call Summarization	Whisper + Qwen2.5	Transcribe and summarize sales calls
Sentiment Analysis	Qwen2.5-0.5B	Analyze customer feedback sentiment
Churn Prediction	XGBoost	Identify customers likely to leave
Next-Best-Action	Qwen2.5 + RAG	Recommend next step for each deal
Customer Segmentation	MiniLM-L6 + Qdrant	Cluster customers by behavior/value
Chatbot	Qwen2.5-3B	Automated customer inquiry handling
Document OCR	PaddleOCR	Extract data from business cards, invoices
Voice CRM	Whisper + Piper TTS	Voice-based CRM for field agents
Multilingual Support	Qwen2.5	Yoruba/Hausa/Igbo customer communications

AI Lead Scoring Architecture

```
Data Sources:
  ├── Website visits (page views, time on site)
  ├── Email engagement (opens, clicks, replies)
  ├── WhatsApp interactions (response time, message count)
  ├── Call logs (duration, frequency)
  ├── Company data (industry, size, location)
  └── Deal history (past purchases, RFQs)
        │
        ▼
  Feature Engineering Pipeline
        │
        ▼
  XGBoost / LightGBM Model
  (trained on historical won/lost deals)
        │
        ▼
  Lead Score (0-100)
  ├── Hot (80-100): Immediate action
  ├── Warm (50-79): Nurture campaign
  └── Cold (0-49): Long-term follow-up
```

AI Impact on CRM Performance

- **77% boost** in lead generation ROI with AI-CRM integration
- **80% increase** in sales productivity
- **25% increase** in sales with predictive analytics
- **30% reduction** in customer churn
- **70% increase** in lead-to-conversion with lead scoring

11. Integration with PECH Platform Stack

Integration Map

PECH Platform	Integration with CRM	Data Flow
Keycloak (Identity)	SSO for CRM users, customer portal auth	User tokens, roles, permissions
ERPNext (ERP)	Quote-to-cash, inventory checks, invoicing	Customers, orders, payments
Medusa (Commerce)	Customer purchase history, wish lists	Orders, products, reviews
Apache Fineract (Fintech)	Customer credit scoring, loan status, PAYG	Payment history, loan applications
ThingsBoard (IoT)	Device ownership, monitoring alerts, usage data	Device IDs, telemetry, alerts
Chatwoot (Support)	Support tickets, live chat history	Tickets, CSAT scores, chat logs
Qdrant (Vector DB)	Semantic search across customer interactions	Embeddings of emails, notes, chats
Apache Superset (BI)	CRM dashboards, reports, analytics	Aggregated CRM data
Apache Airflow (Workflow)	Automated data pipelines, batch processing	ETL jobs, scheduled reports
NATS/Kafka (Events)	Real-time event streaming between systems	Events: lead_created, deal_won, etc.
Apache APISIX (Gateway)	API routing, rate limiting, auth	All API traffic between services

Event Schema Examples

```
// Lead Created Event
{
  "event": "crm.lead.created",
  "timestamp": "2026-03-10T14:30:00Z",
  "data": {
    "lead_id": "LEAD-2026-0042",
    "name": "Adeyemi Ogunlade",
    "company": "Lagos Solar Solutions",
    "source": "whatsapp",
    "domain": "energy",
    "phone": "+2348012345678",
    "assigned_to": "sales-rep-003"
  }
}

// Deal Won Event
{
  "event": "crm.deal.won",
  "timestamp": "2026-03-10T16:45:00Z",
  "data": {
    "deal_id": "DEAL-2026-0018",
    "customer_id": "CUST-0042",
    "amount": 2500000,
    "currency": "NGN",
```

```
    "products": ["solar-panel-5kw", "iot-monitor-basic"],
    "domains": ["energy", "iot"],
    "payment_method": "installment_12m"
  }
}
```

12. CRM Data Model

Core Entities

```
┌──────────────┐      ┌──────────────┐      ┌──────────────┐
│   Contact    │──────→│   Account    │←──────│    Lead      │
│  (Person)    │  N:1 │  (Company)   │  1:N │  (Prospect)  │
└──────────────┘      └──────────────┘      └──────────────┘
       │                    │                      │
       │                    │                      │
       │                ┌──────────┐               │
       └───────────────→│   Deal   │←──────────────┘
                        │(Opportunity)│
                        └──────────┘
                             │
                             │
              ┌──────────────┼──────────────┐
              │              │              │
       ┌──────────┐   ┌──────────┐   ┌──────────┐
       │ Activity │   │ Quotation│   │   Task   │
       │(Call,Email│   │          │   │          │
       │ Meeting) │   │          │   │          │
       └──────────┘   └──────────┘   └──────────┘
```

Key Tables

```
-- Core CRM Tables
contacts (id, first_name, last_name, email, phone, whatsapp, account_id, ...)
accounts (id, name, industry, address_city, address_state, rc_number, ...)
leads (id, name, source, status, score, assigned_to, domain, ...)
deals (id, title, stage, amount, currency, probability, close_date, domain, ...)
activities (id, type, subject, contact_id, deal_id, timestamp, notes, ...)
quotations (id, deal_id, version, total_amount, valid_until, status, ...)
campaigns (id, name, channel, start_date, end_date, budget, roi, ...)
tasks (id, title, assigned_to, due_date, priority, deal_id, status, ...)

-- PECH-Specific Extensions
customer_domains (customer_id, domain, products, status)
  -- Tracks which PECH domains each customer engages with
payg_subscriptions (customer_id, product_id, installment_amount, paid, remaining)
  -- Pay-as-you-go tracking for solar/IoT products
iot_devices (customer_id, device_id, thingsboard_id, status, last_active)
  -- Links CRM customers to their IoT devices
```

13. Implementation Roadmap

Phase 1: Crawl (Months 1-3) — ERPNext CRM

Week	Action	Outcome
1-2	Configure ERPNext CRM module	Lead capture forms, pipeline stages
3-4	Set up lead assignment rules	Auto-assign by territory/domain

5-6	Import existing customer data	Clean, deduplicated customer database
7-8	Connect Chatwoot → ERPNext	Support tickets linked to customers
9-10	Train sales team (5-10 people)	Active CRM usage by sales
11-12	Build basic dashboards in Superset	Pipeline, conversion, revenue reports

Cost: ₦0 additional software (ERPNext already deployed) **Team:** 1 ERPNext admin + sales team **Risk:** Low — using existing, proven system

Phase 2: Walk (Months 4-8) — Add NocoBase + Integrations

Month	Action	Outcome
4	Deploy NocoBase (Docker)	Custom CRM views, no-code dashboards
5	Build cross-domain customer view	360-degree customer profiles
5	WhatsApp integration (Chatwoot)	Automated lead capture from WhatsApp
6	USSD integration (Africa's Talking)	Feature phone customer self-service
6	NATS event bus setup	Real-time CRM ↔ ERP sync
7	AI lead scoring (XGBoost)	Automated lead prioritization
8	Mobile PWA for field agents	Offline-capable mobile CRM

Cost: ~₦2-5M (Africa's Talking USSD, WhatsApp Business API fees) **Team:** 1 developer + 1 CRM admin **Risk:** Medium — new integrations to build and test

Phase 3: Run (Months 9-18) — Full AI-Powered CRM

Month	Action	Outcome
9-10	AI sales forecasting	Predictive revenue models
11-12	AI call summarization (Whisper)	Automated call notes
13-14	Customer churn prediction	Proactive retention campaigns
15-16	Voice CRM (Whisper + Piper TTS)	Voice-based CRM for field agents
17-18	Full Kafka event streaming	Enterprise-grade real-time sync

Cost: ~₦5-10M (compute for AI models, Kafka infrastructure) **Team:** 2 developers + 1 data scientist + 1 CRM admin **Risk:** Higher — AI model training requires clean historical data

---

14. Key CRM Metrics and KPIs

Sales Metrics

KPI	Target	Measurement
Lead Response Time	< 1 hour	Time from lead creation to first contact
Lead-to-Customer Conversion Rate	> 15%	Converted leads / total leads
Sales Cycle Length	< 45 days	Average days from lead to close
Win Rate	> 25%	Won deals / total deals
Average Deal Size	Varies by domain	Total revenue / number of deals
Pipeline Coverage	3x target	Pipeline value / revenue target

Sales Velocity	Increasing	(Deals × Win Rate × Avg Deal Size) / Cycle Length

Customer Metrics

KPI	Target	Measurement
Customer Lifetime Value (CLV)	Increasing	Total revenue per customer over lifetime
Customer Acquisition Cost (CAC)	Decreasing	Marketing + sales cost / new customers
CLV:CAC Ratio	> 3:1	Profitability of customer acquisition
Net Promoter Score (NPS)	> 50	Customer satisfaction survey
Churn Rate	< 5% monthly	Lost customers / total customers
Cross-sell Rate	> 20%	Customers buying from 2+ PECH domains
Repeat Purchase Rate	> 40%	Returning customers / total customers

Operational Metrics

KPI	Target	Measurement
CRM Adoption Rate	> 90%	Active users / total sales team
Data Quality Score	> 85%	Complete records / total records
Activity Per Rep	> 30/week	Calls + emails + meetings per rep
Quote-to-Order Ratio	> 40%	Orders / quotes sent
Support Resolution Time	< 24 hours	Average time to resolve tickets

---

15. NDPA 2023 Compliance

The **Nigeria Data Protection Act 2023 (NDPA)** directly impacts CRM operations. Key requirements:

Compliance Requirements for CRM

NDPA Requirement	CRM Implementation
Lawful basis for processing	Record consent at lead capture; track consent per communication channel
Purpose limitation	Only use customer data for stated purposes; tag data by purpose
Data minimization	Collect only necessary fields; avoid over-collecting
Accuracy	Allow customers to update their records; regular data audits
Storage limitation	Auto-archive inactive records after defined period
Security	Encrypt data at rest (PostgreSQL TDE) and in transit (TLS)
Data subject rights	Enable data export, deletion, and portability on request
Cross-border transfer	Keep CRM data in Nigerian data centers (Lagos co-location)
Breach notification	Automated breach detection and notification workflows
Data Protection Impact Assessment	Document CRM's data processing activities

Technical Implementation

1. **Consent management:** CRM must record when, how, and for what purpose consent was given

2. **Right to erasure:** Ability to permanently delete a customer record and all associated data
3. **Data portability:** Export customer data in standard format (JSON, CSV)
4. **Audit trail:** Log all access to and modifications of customer data
5. **Encryption:** AES-256 for data at rest, TLS 1.3 for data in transit
6. **Access control:** Role-based access control (RBAC) via Keycloak
7. **Data residency:** All CRM data stored in Lagos data center

---

16. Cost Analysis

Self-Hosted CRM Cost Breakdown (Lagos, Nigeria)

Phase 1: ERPNext CRM (₦0 Additional)

Component	Cost	Notes
ERPNext CRM module	₦0	Already deployed
Configuration & setup	₦500K	Internal team time (2 weeks)
Data migration	₦300K	Import existing customer data
Training	₦200K	2-day workshop for sales team
Phase 1 Total	**₦1M**	~$640 USD

Phase 2: NocoBase + Integrations (₦5-8M)

Component	Cost/Month	Annual Cost	Notes
NocoBase server (Docker)	₦0	₦0	Runs on existing TRX50
Africa's Talking USSD	₦150K	₦1.8M	USSD sessions at ~₦3/session
WhatsApp Business API	₦100K	₦1.2M	Meta pricing + Chatwoot
Developer time	₦300K	₦3.6M	1 part-time developer
Hosting (additional compute)	₦50K	₦600K	Extra containers on TRX50
Phase 2 Annual		**₦7.2M**	~$4,600 USD

Phase 3: AI-Powered CRM (₦10-15M)

Component	Cost/Month	Annual Cost	Notes
AI model compute (RTX 4090)	₦0	₦0	Shared GPU, already budgeted
Kafka cluster	₦100K	₦1.2M	3-node cluster for event streaming
Additional developer	₦500K	₦6M	Full-time CRM/AI developer
Data scientist	₦400K	₦4.8M	ML model development
Infrastructure scaling	₦100K	₦1.2M	Additional compute/storage
Phase 3 Annual		**₦13.2M**	~$8,500 USD

Cost Comparison: Self-Hosted vs SaaS

Solution	20 Users/Year	50 Users/Year	Notes
PECH Self-Hosted (ERPNext + NocoBase)	₦7.2M (~$4.6K)	₦7.2M (~$4.6K)	Flat cost, unlimited users
Salesforce	₦18.7M (~$12K)	₦46.8M (~$30K)	$25-300/user/month

HubSpot (Professional)	₦14M (~$9K)	₦14M (~$9K)	Flat rate but limited
Zoho CRM (Professional)	₦4.7M (~$3K)	₦11.7M (~$7.5K)	$23/user/month

Self-hosted wins at scale. At 50+ users, PECH saves 60-85% vs SaaS alternatives. With unlimited users and full data ownership, the ROI improves every year.

---

17. Sources

CRM Market & Statistics

- CRM Statistics 2026 — Cyntexa
- CRM Market Size to Hit $304B by 2035 — Precedence Research
- CRM Market Forecast 2025-2029 — Technavio
- Nigeria CRM Software Market — Statista
- Africa CRM Software Market — Statista
- Zoho 70% YoY Growth in Nigeria — TechCabal
- CRM Statistics 2026 — DemandSage

Open-Source CRM Platforms

- NocoBase — Apache-2.0 No-Code Platform
- NocoBase CRM Comparison — DEV Community
- BottleCRM — MIT Licensed CRM
- Django-CRM GitHub — MicroPyramid
- Krayin CRM — MIT Licensed Laravel CRM
- Krayin CRM GitHub
- Top 20 Open-Source CRMs — Grow CRM
- Open Source CRM Benchmark 2025 — Marmelab
- Best Open Source CRM 2026 — CRM.org

ERPNext CRM

- ERPNext CRM System — Sigzen
- ERPNext CRM Guide — Frappe
- ERPNext Modules for 2025 — TechClouds
- Frappe CRM GitHub

CRM-ERP Integration

- CRM-ERP Integration Best Practices — Kogifi
- ERP Integration Guide 2025 — DCKAP
- CRM Microservices Architecture — DigiQT
- Microservices in ERP — TechWize

Africa Mobile & USSD

- WhatsApp Integration for Africa — HelloDuty
- Africa's Talking USSD API
- Celcom Africa USSD Services
- CRM Africa — Best CRM for Small Business

AI in CRM

- Top 10 Open Source AI CRM Tools — SuperAGI

- AI Lead Scoring Guide — Reform
- AI Lead Scoring — Warmly